	AIR UNIVERSITY
	DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
	EXPERIMENT NO 13

Lab Title Fourier Transform & GUI

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Objective:

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks: _____

Date: _____ Signature: _____



Department of Electrical and Computer Engineering

Operating Systems Lab

Lab Report: 13

Title: Fourier Transform

& GUI

Name: Ali Abbas Khan

Roll Number: 230661

Class: BCE-5B

Submitted To: Engr. MAryam
Date: December 30, 202

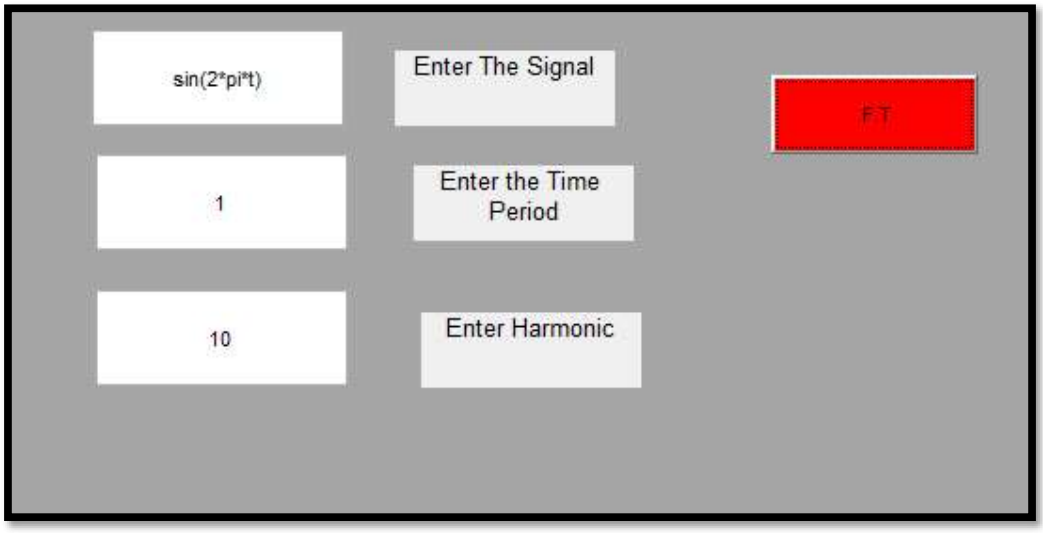
Task 1:

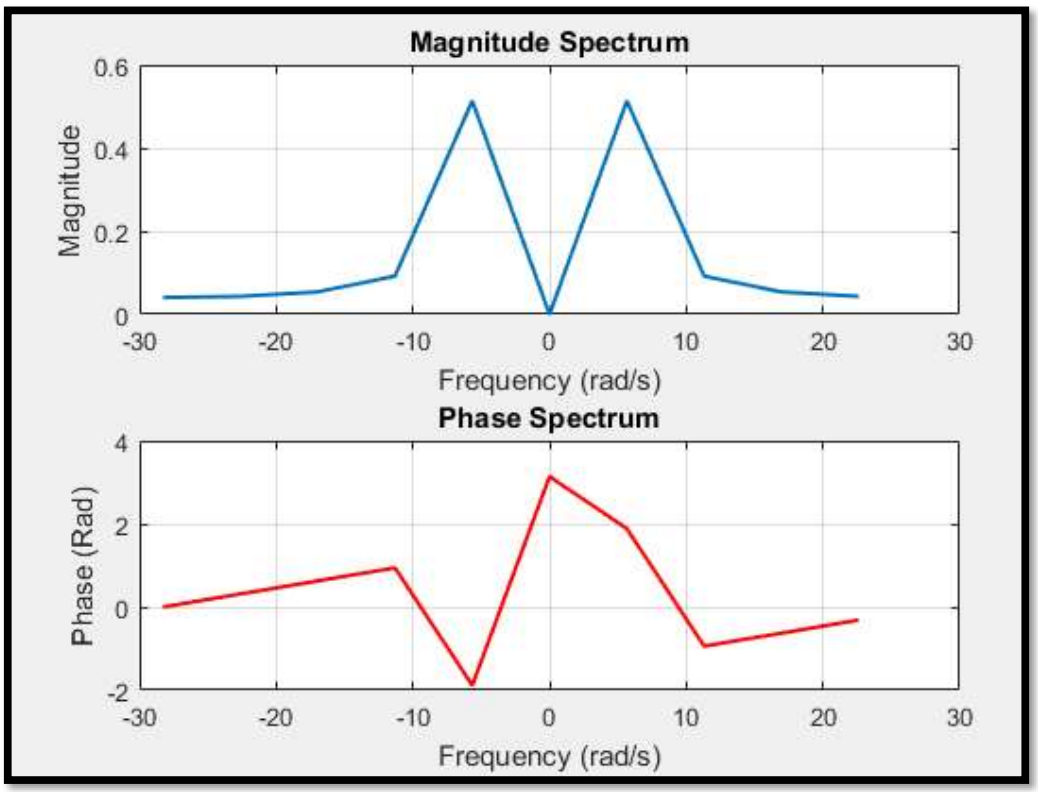
Make a GUI of that takes signal harmonics and Time period as input and gives us Plot of its magnitude and phase spectrum.

Code:

```
147     % hObject    handle to pushbutton1 (see GCBO)
148     % eventdata  reserved - to be defined in a future version of MATLAB
149     % handles    structure with handles and user data (see GUIDATA)
150 -   expr=get(handles.edit4,'string');
151 -   T=str2double(get(handles.edit5,'string'));
152 -   N=str2double(get(handles.edit6,'string'));
153
154 -   t=linspace(-T/2,T/2,N);
155 -   dt=t(2)-t(1);
156
157 -   x_t=eval(vectorize(expr));
158
159 -   X_w=fftshift(fft(x_t))*dt;
160 -   f=(-N/2:N/2-1)/(N*dt);
161 -   w=2*pi*f;
162
163 -   figure;
164 -   subplot(2,1,1);
165 -   plot(w,abs(X_w),'LineWidth',1.5);
166 -   title('Magnitude Spectrum');
167 -   xlabel('Frequency (rad/s)');
168 -   ylabel('Magnitude');
169 -   grid on;
170
171 -   subplot(2,1,2);
172 -   plot(w,angle(X_w),'r','LineWidth',1.5);
173 -   title('Phase Spectrum');
174 -   xlabel('Frequency (rad/s)');
175 -   ylabel('Phase (Rad)');
176 -   grid on;
```

Output:





Conclusion:

In this lab, we successfully designed a MATLAB GUI that allows users to input a time-domain signal expression, specify the observation interval, and set the number of samples to compute its Fourier spectrum. By leveraging FFT and proper scaling, the program generates both the magnitude and phase spectra, providing clear visualization of the signal's frequency content. This exercise reinforced key concepts of signal analysis, including sampling, discretization, and spectral representation, while also demonstrating how interactive GUIs can make theoretical ideas more intuitive and practical.